Assessing Studies Based on Multiple Regression

SW Chapter 9

ECON3500: Econometrics and Applications

Spring 2026

Learning Objectives

By the end of this chapter, you will be able to:

Define **internal validity** and **external validity**

Identify five threats to internal validity

Explain how each threat causes bias or incorrect inference

Propose solutions to each threat

Distinguish between **causal inference** and **forecasting** objectives

Aside: Linear Probability Models

What if the **dependent variable** is binary (0/1)?

$$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + u_i$$

If y only takes values 0 and 1: $E(y|\mathbf{x}) = P(y = 1|\mathbf{x}) = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k$

$\Rightarrow$ Coefficients measure the **change in probability** that $y = 1$ for a one-unit change in x_j

Limitations: predicted probabilities can fall outside $[0, 1]$; necessarily heteroskedastic (always use `robust` SEs)

Advantages: easy to estimate and interpret; works naturally with multiple regression tools (controls, IV, fixed effects)

When Is the LPM Appropriate?

The LPM works well when:

Predicted probabilities are not very high or very low (no strict rule)

You care about **average marginal effects**, not predicting exact probabilities

Your regressors are mostly **categorical** (dummy variables, interactions)

The LPM is **less appropriate** when:

Many predicted probabilities are near 0 or 1 (e.g., rare events)

You need the predicted probabilities themselves (e.g., for classification or risk scoring)

Bottom line: - For **causal inference** — which is our focus — the LPM is a perfectly good tool. - Angrist & Pischke (*Mostly Harmless Econometrics*) argue it gives you the right average effects without the complexity of nonlinear models. - More formal alternatives (probit, logit) exist but are beyond ECON3500. - **You will use the LPM in Lab 6!**

Internal and External Validity

What Is Validity?

Validity = Do we believe the results of our estimation?

Did we learn what we set out to learn?

Did we estimate the causal impact of class size on test scores?

Did we correctly measure the association between marital status and wages?

We need a **framework** for evaluating whether a study is useful for answering a particular question.

What are the **threats to validity**?

Two Dimensions of Validity

Internal Validity

The statistical inferences about causal effects are valid **for the population being studied**.

External Validity

The statistical inferences can be **generalized** from the population and setting studied to other populations and settings.

"Setting" = the legal, policy, and physical environment and related salient features.

External Validity Example: Seattle Minimum Wage

Question: A study finds Seattle's $15 minimum wage **reduced hours** for low-wage workers (Jardim et al. 2022, *AEJ: Economic Policy*). Would we expect the same result elsewhere?

American Economic Journal: Economic Policy 2022, 14(2): 263–314
https://doi.org/10.1257/pol.20180578

Minimum-Wage Increases and Low-Wage Employment: Evidence from Seattle†

By Ekaterina Jardim, Mark C. Long, Robert Plotnick, Emma van Inwegen, Jacob Vigdor, and Hilary Wething*

Seattle raised its minimum wage to as much as $11 in 2015 and as much as $13 in 2016. We use Washington State administrative data to conduct two complementary analyses of its impact. Relative to outlying regions of the state identified by the synthetic control method, aggregate employment at wages less than twice the original minimum—measured by total hours worked—declined. A portion of this reduction reflects jobs transitioning to wages above the threshold; the aggregate analysis likely overstates employment effects. Longitudinal analysis of individual Seattle workers matched to counterparts in outlying regions reveals no change in the probability of continued employment but significant reductions in hours, particularly for less experienced workers. Job turnover declined, as did hiring of new

Jardim et al. (2022), *AEJ: Economic Policy*

Threats to External Validity

Assessing external validity requires **detailed substantive knowledge** and judgment on a case-by-case basis.

Same example: Seattle's $15 minimum wage and hours worked. What should we ask before generalizing?

Differences in populations

- Seattle's low-wage workforce: tech-adjacent, high cost of living
- Rural Vermont? Mississippi? Workers face very different outside options

Differences in settings

- Seattle's booming economy may have cushioned the shock
- Different state labor laws, tipped wage rules, enforcement

Differences in labor markets

- Urban labor markets are less concentrated — more employers competing for workers
- Research shows more concentrated (rural) labor markets may respond **differently** to minimum wage increases (Azar et al. 2023)

There is no mechanical test for external validity — it requires **judgment**.

Internal Validity: Two Requirements

Estimates are internally valid if:

The estimator is **consistent** (or unbiased)

Confidence intervals have **correct coverage** — in at least 95% of samples, the 95% CI contains the true parameter

The Rest of This Chapter

We focus on threats to **internal validity**: things that can make our regression estimates misleading for the population we're studying.

Five Threats to Internal Validity

Overview

Five threats to the internal validity of regression studies:

Omitted variable bias

Wrong functional form

Errors-in-variables bias (measurement error)

Sample selection bias

Simultaneous causality bias

All five imply that $E(u_i|X_{1i},\ldots,X_{ki}) \neq 0$

$\Rightarrow$ Conditional mean independence fails

$\Rightarrow$ OLS is **biased and inconsistent**

> ⓘ **Quick Review**
>
> **Biased**: $E[\hat{\beta}] \neq \beta$ — on average across samples, our estimator gets the wrong answer
>
> **Consistent**: as $n \to \infty$, $\hat{\beta}$ converges in probability to β — more data gets you closer to the truth. When an estimator is **inconsistent**, more data doesn't fix the problem.

Three Categories of Threats

When you encounter a potential threat, classify it:

Category	What goes wrong	Severity
1. Affects **consistency** of $\hat{\beta}$	Estimator converges to wrong value	Most serious
2. Affects **inference** but not consistency	CIs have wrong coverage; $\hat{\beta}$ is still consistent	Serious
3. Increases **imprecision** only	Larger SEs, but $\hat{\beta}$ unbiased and CIs correct	Least serious

Categories 1 and 2 are **actual threats** to validity

Category 3 makes estimates imprecise but standard errors properly reflect that imprecision

1. Omitted Variable Bias

We know this one! OVB arises if an omitted variable is both (1) a **determinant of Y**, and (2) **correlated with** at least one included regressor.

With control variables included, are there still omitted factors that are not adequately controlled for? Is the error term correlated with the variable of interest even after we have included the control variables?

Solutions:

If the omitted variable **can be measured**: include it as an additional regressor

If you have **adequate controls** (conditional mean independence plausibly holds): include them

Run a **randomized controlled experiment** — if X is randomly assigned, $E(u|X) = 0$

Use **panel data** (Ch10) or **instrumental variables** (Ch12) — coming soon!

2. Wrong Functional Form

Arises if the functional form is incorrect — for example, an interaction term is incorrectly omitted; then inference on causal effects will be biased.

Solutions to functional form misspecification:

Continuous Y: use the appropriate nonlinear specifications in X (logarithms, interactions, etc. — Ch8)

Discrete Y (e.g., binary): use a **linear probability model** (as in Lab 6), or more formally, probit or logit (beyond ECON3500)

Note: This issue is rarely our biggest problem in regression analysis.

3. Errors-in-Variables Bias

So far we've assumed X is measured without error. In reality, economic data often have measurement error:

Data entry errors in administrative data

Recollection errors in surveys ("When did you start your current job?")

Ambiguous questions ("What was your income last year?" — before or after tax?)

Intentionally false responses ("What is the current value of your financial assets?" "How often do you drink and drive?")

Example: Survey Responses as Measurement Error

What kind of measurement error might this create?

> But a new research paper points out one huge potential flaw in all this research: kids who skew the results by making stuff up for a giggle. "Mischievous Responders," they're called.
>
> They may say they're 7 feet tall, or weigh 400 pounds, or have three children. They may exaggerate their sexual experiences, or lie about their supposed criminal activities. In other words, kids will be kids, especially when you ask them about sensitive issues.
>
> Jackson Terry, 14, says he answered honestly when he took one of these surveys last year, but he knows kids who didn't.
>
> "They handed out the sheet, I believe it was in language class," says Terry, who's from Granville, Ohio. "The teacher was in the room. It was anonymous. I think they asked us about bullying, do you feel safe in school, some questions about drugs, the learning environment."
>
> Some kids "would joke through the entire thing and have a cocky attitude about it," Terry says. "Afterwards some would say, yeah, No. 5, that's totally not true; I just made something up."

Example of potentially inaccurate self-reported survey responses

Self-reported survey data can be wrong because respondents **misremember**, **misunderstand**, or **make things up**

For sensitive questions, the error may be **systematic**, not purely random

That means measurement error in survey data is often **not classical**

Where and What Kind?

Measurement error can occur in the **dependent variable** or **independent variables**.

And it can be **classical** (random) or **non-classical** (systematic).

	Classical (random)	Non-classical (systematic)
Dependent variable	Minor problem — larger variance	Serious problem — bias
Independent variable	**Attenuation bias**	Serious problem — bias

Classical Errors-in-Variables (CEV) Assumption

The measurement error is uncorrelated with the true value: $\mathrm{Cov}(w_i, X_i^*) = 0$

Measurement Error in Y

Suppose we observe y_i but the true value is y_i^*:

$$y_i = y_i^* + e_i$$

The population regression is:

$$y_i^* = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u_i$$

What we actually estimate:

$$y_i = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + \underbrace{(u_i + e_i)}_{\text{new error term}}$$

If e_i is uncorrelated with the X's, the **coefficients are still unbiased** — the error term just has more variance, so standard errors are larger.

Measurement error in Y under CEV: not a big deal.

Measurement Error in X: Setup

The true model is:

$$Y_i = \beta_0 + \beta_1 X_i^* + u_i$$

We observe $X_i = X_i^* + w_i$ instead of X_i^*, where $E[w_i] = 0$.

Assumptions (classical errors-in-variables):

$\mathrm{Cov}(w_i, u_i) = 0$ — measurement error uncorrelated with other omitted factors

$\mathrm{Cov}(w_i, X_i^*) = 0$ — measurement error uncorrelated with the true value $\leftarrow$ **the CEV assumption**

$\mathrm{Cov}(X_i^*, u_i) = 0$ — no traditional OVB

Why Is This a Problem?

The key insight: even though $\mathrm{Cov}(w_i, X_i^*) = 0$, the measurement error w_i **is** correlated with the observed X_i.

$$\mathrm{Cov}(X_i, w_i) = \mathrm{Cov}(X_i^* + w_i,\ w_i)$$

$$= \underbrace{\mathrm{Cov}(X_i^*, w_i)}_{=\ 0 \text{ by CEV}} + \mathrm{Cov}(w_i, w_i)$$

$$= \mathrm{Var}(w_i) = \sigma_w^2$$

The observed regressor X_i is correlated with the measurement error, which is part of the composite error term. This **violates conditional mean independence**.

Formal Derivation of Measurement Error (1/2)

$$\hat{\beta_1} = \frac{\text{Cov}(X_i, Y_i)}{\text{Var}(X_i)}$$

$$= \frac{\text{Cov}(X_i^* + w_i,\ \beta_0 + \beta_1 X^* + u_i)}{\text{Var}(X_i^* + w_i)}$$

Expanding the numerator:

$$= \frac{\text{Cov}(X_i^*, \beta_0) + \text{Cov}(X_i^*, \beta_1 X_i^*) + \text{Cov}(w_i, \beta_0) + \text{Cov}(w_i, \beta_1 X^*) + \text{Cov}(w_i, u_i)}{\text{Var}(X_i^*) + \text{Var}(w_i) + 2\text{Cov}(X_i^*, w_i)}$$

Formal Derivation of Measurement Error (2/2)

Applying our assumptions ($\text{Cov}(w, X^*) = 0$, $\text{Cov}(w, u) = 0$):

$$= \frac{0 + \beta_1 \text{Var}(X_i^*) + 0 + 0 + 0}{\text{Var}(X_i^*) + \text{Var}(w_i) + 0}$$

$$= \beta_1 \cdot \frac{\text{Var}(X_i^*)}{\text{Var}(X_i^*) + \text{Var}(w_i)}$$

Attenuation Bias: The Result

$$\hat{\beta_1} = \beta_1 \cdot \frac{\text{Var}(X_i^*)}{\text{Var}(X_i^*) + \text{Var}(w_i)}$$

The fraction on the right is **between 0 and 1**. So:

If $\beta_1 > 0$: estimate is biased **toward zero** (too small)

If $\beta_1 < 0$: estimate is biased **toward zero** (too close to zero)

> **Attenuation Bias**
>
> Classical measurement error in a regressor always biases the coefficient **toward zero** — making the estimated effect look **smaller** than it truly is.

Attenuation Bias: Intuition

Why does this happen?

X_i has a **higher variance** than the true X_i^* (the noise adds spread)

When X_i moves, Y doesn't always respond — because Y only responds to movements in X^*

The regression sees changes in X that are **not related to** Y

It concludes: "the effect of X on Y must be small"

Practical implication: If we think there's measurement error in X, our estimate is a **lower bound** on the true magnitude.

What If CEV Doesn't Hold?

Sometimes the classical assumption is unlikely:

People with high incomes **under-report** and people with low incomes over-report

- $\Rightarrow \text{Cov}(\text{income}, \text{error}) \neq 0$

People are less accurate about their age as they get older

- $\Rightarrow \text{Cov}(\text{age}, \text{error}) \neq 0$

When CEV fails:

The direction of bias is **ambiguous** — no longer necessarily toward zero

Interpreting the impact is more complicated (beyond ECON3500)

Summary of Measurement Error

Dependent variable (classical): Don't worry too much — OLS still BLUE, just larger variance

Independent variable (classical): OLS no longer BLUE — **attenuation bias**

- We can still sign our magnitudes (e.g., "Returns to education are **at least** 10%")

Independent variable (non-classical): OLS no longer BLUE — bias in ambiguous direction

Better data always helps! Use administrative records instead of survey self-reports when possible.

Use **instrumental variables** (later)

4. Missing Data and Sample Selection Bias

Data are often missing. Whether this introduces bias depends on **why** the data are missing:

Case	Data missing because of...	Bias?
1	**Random chance**	No
2	**Values of X**	No
3	**Values of Y or u**	**Yes**

Cases 1 and 2: standard errors are larger, but $\hat{\beta}$ is **unbiased**

Case 3: introduces **sample selection bias**

Case 1: Missing at Random

Suppose you survey 100 workers and record answers on paper — but your dog eats 20 response sheets (selected at random) before you enter them.

This is equivalent to having a random sample of 80 workers.

Your dog didn't introduce any bias!

Knowledge Check

A server crash deletes 30% of your dataset at random. Should you worry about bias?

No — this is Case 1. You lose precision (larger SEs) but no bias.

Case 2: Missing Based on X

Suppose you're studying the effect of education on wages, but you restrict your sample to **workers under age 30**.

You can't say anything about older workers

But restricting to younger workers **doesn't introduce bias** for the under-30 population

This is equivalent to having missing data where data are missing based on X (age)

More generally: if data are missing based **only on values of X**, the missing data don't bias OLS. (It does limit **external validity** — you can't generalize beyond that subgroup.)

Case 3: Missing Based on Y or u

This **does** introduce bias. This is **sample selection bias**.

Sample selection bias arises when a selection process:

Influences the **availability of data**, and

Is **related to the dependent variable**

In DAG terms: this is **collider bias** — conditioning on a variable that is caused by both X and Y (recall Ch 8b slides).

> **The Key Question**
>
> Ask yourself: "Would the **reason** someone is missing from my data be correlated with the **outcome** I'm trying to measure?" If yes $\longrightarrow$ sample selection bias.

Example: Height of Undergraduates

You want to estimate the mean height of undergraduates.

You collect data by standing outside the **basketball team's locker room** and recording the height of undergraduates who enter.

Is this a good design?

No

You've sampled individuals in a way that is **related to the outcome** Y (height)

This results in bias — your estimate will be too high

Example: Mutual Fund Performance

Question: Do actively managed mutual funds outperform "hold-the-market" index funds?

Design:

Sample: mutual funds available to the public **today**

Data: returns for the preceding 10 years

Compare: average 10-year return vs. S&P 500

Is there sample selection bias?

Yes — funds that performed poorly **went out of business** and are not in today's sample.

The surviving funds are the "basketball players" of mutual funds.

This is called **survivorship bias** — the surviving funds are the "basketball players" of mutual funds.

Being managed *and in our sample* means the fund outperformed failed managed funds $\Rightarrow \mathrm{Cov}(\mathrm{managedfund}_i, u_i) \neq 0 \Rightarrow$ **OLS is biased**

Example: Returns to Education

Question: What is the return to an additional year of education among college graduates?

Design:

Sample: **employed** college graduates (so we have wage data)

Regress: $\ln(\text{earnings})$ on years of education

Is there sample selection bias?

Yes — we only observe wages for people who are **employed**. Employment is related to the outcome (wages/earnings potential), so the sample is not representative.

Solutions to Sample Selection Bias

Collect data properly — design sampling to avoid selection

- *Basketball:* true random sample from enrollment records
- *Mutual funds:* sample funds available at the **beginning** of the period (include failed funds)
- *Returns to education:* sample college graduates, not just employed workers

Randomized controlled experiment

Model the selection process and estimate it (advanced — not in ECON3500)

5. Simultaneous Causality Bias

So far we've assumed X causes Y. But what if Y **also causes** X?

This is **simultaneous causality** (or "reverse causality"). Remember: the "A" in DAG stands for **acyclic** — DAGs **cannot represent** simultaneity (Ch 8b slides).

Example: Class size effect

Low STR $\longrightarrow$ better test scores (the effect we want)

But: districts with **low test scores** may receive extra resources $\longrightarrow$ also get low STR (political process)

What does this mean for a regression of TestScore on STR?

The coefficient on STR reflects **both** directions of causality — it's biased.

Solutions to Simultaneous Causality Bias

Randomized controlled experiment — because X_i is chosen at random, there is no feedback from Y to X (assuming perfect compliance)

Develop a complete model of both directions of causality — this is the approach behind large macroeconomic models (e.g., Federal Reserve). **Extremely difficult in practice.**

Instrumental variables regression — estimate the causal effect of X on Y while ignoring the feedback from Y to X (coming later!)

Summary: Five Threats to Internal Validity

Threat	What Happens	Consequence	Solutions
1. **OVB**	Omitted variable correlated with X and determines Y	Biased, inconsistent	Add controls, RCT, IV, panel data
2. **Wrong functional form**	Incorrect specification (missing interactions, nonlinearities)	Biased	Logs, quadratics, interactions (Ch8)
3. **Measurement error**	X or Y measured with error	Attenuation bias (classical); ambiguous bias (non-classical)	Better data, IV
4. **Sample selection**	Data missing based on Y or u	Biased, inconsistent	Better sampling design, RCT
5. **Simultaneous causality**	Y causes X and X causes Y	Biased, inconsistent	RCT, IV

Forecasting vs. Causal Inference

Forecasting vs. Causal Inference

	Causal Inference	Forecasting
Goal	Estimate effect of changing X on Y	Predict Y given observed X's
What matters most	Unbiased $\hat{\beta}$	Good fit ($\bar{R}^2$)
OVB	**Critical problem**	Not a problem!
Coefficient interpretation	Very important	Not important
External validity	Important for generalization	**Paramount** — model must hold in new data

⚠ **Don't Confuse the Two**

A model with high R^2 and biased coefficients might be great for forecasting but useless for policy. A model with unbiased coefficients and low R^2 might be ideal for causal inference but poor for prediction.

Knowledge Check: Identify the Threat

For each scenario, identify the primary threat to internal validity:

1. A study of returns to education uses IQ as a proxy for ability, but IQ tests are noisy measures of true ability.

$\longrightarrow$ **Errors-in-variables bias** (measurement error in X) — attenuation bias on the IQ coefficient

2. A study of the effect of police on crime finds that cities with more police have more crime.

$\longrightarrow$ **Simultaneous causality** — high-crime cities hire more police

3. A study of job training effects on wages only observes wages for people who are employed after training.

$\longrightarrow$ **Sample selection bias** — employment status is related to the outcome (wages)

Key Takeaways

Measurement error in X (classical) causes attenuation bias — estimates are too small in magnitude

Sample selection bias arises when data are missing because of Y, not because of X or random chance

Simultaneous causality is hard to solve without experiments or instrumental variables

Looking Ahead

Panel data (Ch10) and **instrumental variables** (Ch12) are direct responses to OVB and simultaneous causality.